

A Density Extrapolation Method for Noncubic Equations of State

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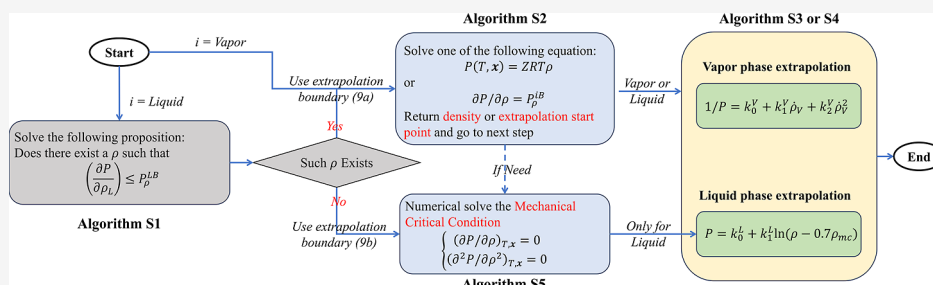
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ABSTRACT: Fluid phase equilibrium calculations are essential for determining system thermodynamic states and underpin chemical process simulations. Iterative calculations in infeasible phase regions may lead to trivial solutions or convergence failure. Existing extrapolation strategies are mostly limited to cubic equations of state (CEoS) and rely on analytical mechanical critical points, restricting their application to higher-order, nonanalytical EoS such as PC-SAFT. To address this, we introduce a general density extrapolation method applicable to EoS without closed-form solutions. The method employs robust numerical techniques to determine the starting point of density extrapolation and the mechanical critical point, thereby developing a more stable approach for vapor phase density extrapolation to estimate densities in infeasible phase regions. Validation through several phase equilibrium cases using the PC-SAFT EoS demonstrates that this method is efficient in terms of convergence and robust for nonlinear and noncubic EoS.

1. INTRODUCTION

The calculation of fluid phase equilibrium,^{1,2} bubble point, and dew point is essential for process simulation³ and is the foundation and computational core of many chemical engineering applications.⁴ In a common approach for the vapor–liquid equilibrium (VLE) calculation, the vapor phase has been described using an EoS, while the liquid phase has typically been represented using activity coefficient models. However, in the past decades, the applicability of EoS has been improved. Noncubic EoS, such as the perturbed-chain statistical associating fluid theory⁵ (PC-SAFT) and its various extensions,^{6–8} have gained prominence due to excellent predictive performance across a wide range of systems, including polymers, electrolytes, and ionic liquids.⁹ Modeling both phases within a unified EoS framework has gained increasing popularity in industrial thermodynamic applications,^{10–12} particularly for high-pressure systems, due to the improved consistency across a broad range of conditions. However, it also introduces numerical challenges, most notably, the risk of convergence to trivial solutions, where both phases yield identical densities and compositions.^{12–14} This problem is especially prevalent in bubble point and dew point calculations,^{15,16} where the absence of fixed temperature or pressure values can cause the phase equilibrium constant K_i to converge to a nonphysical solution after a few iterations over a broad range.

Category strategies to avoid “trivial solutions”^{17–25} are divided into two primary types. The first type involves modifications or reformulations at the upper-algorithm level, which refers to applications that rely on the volumetric properties provided by the thermodynamic model (e.g., single-stage flash calculations). The second type employs the use of pseudoroots at the lower (thermodynamic model) level when solving for density or volume. Following early methods such as that of Asselineau et al.,¹⁷ who proposed a reinitialization method based on pseudocritical properties and a damping technique to stabilize iterations. However, when the density iteration approaches the single-phase region, reinitialization becomes computationally expensive. Pasad et al.¹⁹ employed second-order thermophysical properties as criteria for identifying trivial vapor and liquid roots while adaptively adjusting vapor phase fractions to prevent convergence to trivial solutions. More studies have focused on pseudodensity (or pseudovolume) methods^{20–27} for single-phase calculations. These methods generate pseudodensities or pseudovolumes in

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the infeasible phase region to replace the EoS roots, thereby avoiding trivial solutions and facilitating the convergence of upper-level phase equilibrium calculations. The generation of pseudodensities requires careful consideration of their expressions and applicable ranges.

Poling¹⁸ and colleagues developed a heuristic density judgment criterion, defining isothermal compressibility (eq 1) to distinguish phase densities. When β satisfies eq 2, it is identified as a liquid-like density, whereas when β satisfies eq 3, it is identified as a vapor-like density

$$\beta = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T \quad (1)$$

$$\beta < \frac{0.005}{101325} \text{ Pa}^{-1} \quad (2)$$

$$\frac{0.9}{P} < \beta < \frac{3}{P} \quad (3)$$

where P and V are the system pressure and volume, respectively. Although the Poling criterion is simple and efficient, it could fail in some cases.²² Bennet and Schmidt²⁸ tested five phase identification methods on different fluids, and the results indicated that the isothermal compressibility method, thermal expansion coefficient method, and negative flash vaporization method performed better than the pseudocritical point method and the volume method. Mill et al.²⁷ proposed a method using extrema (stationary points) on the P – ρ curve predicted by the BWRS equation to determine the density of nonexistent phases. Similar approaches are found in Gunderson²⁶ and Lee.²⁰ A key limitation of these methods arises under mechanically supercritical conditions ($T > T_{\text{mc}}$), where the absence of extrema on the P – ρ curve prevents the effective separation of vapor-like and liquid-like densities. Lucia and Taylor²⁹ extended the compressibility factor and flash calculation into the complex domain, where pseudoproperties of phases are computed, effectively avoiding trivial solutions.

Mathias et al.²¹ initially proposed an extrapolation method based on mechanical critical conditions (where both the first and second derivatives of pressure with respect to density are zero) to replace the infeasible phase roots in the EoS. Zhao and Saha²² built upon the complex-domain framework introduced by Lucia,²⁹ extending it to address the loss of phase continuity properties. By representing the EoS roots as complex numbers with liquid-like or vapor-like attributes, they constructed pseudoroot using the extrapolated root rather than EoS stationary points. Mei et al.²³ improved the extrapolation points in the Zhao–Saha method²² to enhance the continuity and smoothness of the pseudoroot. Stateva et al.²⁴ performed extrapolation in different regions of the P – v coordinate system, using second- or third-order spline functions for extrapolation. Recently, Watson and Barton²⁵ employed a nonsmooth median function to modify the original extrapolation formulation of Mathias et al.,²¹ addressing the issue of missing sensitivity information during extrapolation calculations.

For the root identification problem in the equation-oriented (EO) method, Kamath et al.³⁰ proposed a criterion to distinguish vapor and liquid solutions in CEoS and reformulated the phase equilibrium problem to derive first- and second-order derivative constraints for both phases. They constructed the objective function using a penalty term and applied complementary variable relaxation to enforce the constraints on the phase equilibrium and vapor–liquid

solution sets. Building on this framework, Dowling et al.³¹ introduced a classification scheme that divides temperature–pressure conditions into four quadrants based on the critical point. In the quadrant where $P > \bar{P}$, where $\bar{P}$ represents the critical pressure, single-phase constraints were implemented using relaxation variables and complementary constraints. Additionally, by using relaxation variables, single-phase solutions were controlled to stay below the bubble point and above the dew point, avoiding nonphysical single-phase solutions in the two-phase region.

Furthermore, a category of methods^{32–35} that avoids inverting EoS (calculate the density from pressure and temperature) is applicable for tracing phase equilibrium isopleths. These methods are highly suitable for multiroot, noncubic EoS, as only solving for density at the initial point. However, their reliance on continuation-type techniques (e.g., extrapolation or integration) renders them unsuitable for applications such as process simulations, where consecutive equilibrium states are not guaranteed to be close.

Compared to the fixed-point pseudoroot or single-phase criteria, the extrapolated pseudoroot method is applicable over a wider temperature range, and the constructed isotherms are continuous and smooth, without the need to introduce the complex domain. However, existing extrapolated pseudoroot methods are based on mechanical critical conditions, which are advantageous for CEoS models, as analytical solutions of mechanical critical conditions can be derived. For higher-order EoS, such as the PC-SAFT equation, where mechanical critical conditions cannot be solved analytically, a stable numerical solution is required. This work aims to develop a more general and robust extrapolation method for higher-order EoS.

2. BACKGROUND

This section briefly describes mechanical stability and introduces the variations of density with pressure and temperature in EoS. It also outlines the characteristics of the existing methods for extrapolating density.

2.1. Mechanical Stability Limit. For a c component system with temperature T and composition x_i , points that satisfy the following conditions are termed mechanical stability limit points

$$\left(\frac{\partial P}{\partial \rho} \right)_{T, x_i} = 0, i = 1, \dots, c \quad (4)$$

As the temperature changes, the mechanical stability limit describes an envelope curve. For a pure fluid ($c = 1$), this curve is the spinodal. However, for a multicomponent mixture ($c > 1$), the locus from eq 4 defines a “mechanical instability” curve, which is typically enclosed by the spinodal curve. The vertex of the mechanical instability curve corresponds to the mechanical critical point, which, in addition to satisfying eq 4, also requires that the second-order partial derivatives be zero

$$\left(\frac{\partial^2 P}{\partial \rho^2} \right)_{T, x_i} = \left(\frac{\partial P}{\partial \rho} \right)_{T, x_i} = 0, i = 1, \dots, c \quad (5)$$

It is noteworthy that for pure substances, the mechanical critical point coincides with the true critical point. In contrast, the mechanical critical point of a mixture lacks physical significance, as it arises from applying the stability criterion of pure substances to mixtures. The mechanical critical density of a mixture (ρ_{mc}) is generally slightly less than or equal to the

true critical density (ρ_c), with equality holding at a critical azeotrope. Compared with the spinodal curve and the critical point, the mechanical stability limit points and the mechanical critical point are easier to determine. The spinodal curve can be defined by the thermodynamic stability criterion, which can be found in [Supporting Information](#).

Accordingly, the mechanical stability limit is used as a criterion for evaluating the stability of EoS roots, while the mechanical critical point serves as a reference for distinguishing liquid-like and vapor-like roots. For representation convenience, the region below it is referred to as mechanically subcritical, and the region above it is referred to as mechanically supercritical.

2.2. Density Variation with Pressure and Temperature in EoS. Most EoS exhibit a mechanically subcritical form within a certain temperature range, such as the 400–454 K isotherms shown in [Figure 1](#), where 1–3 roots may exist for

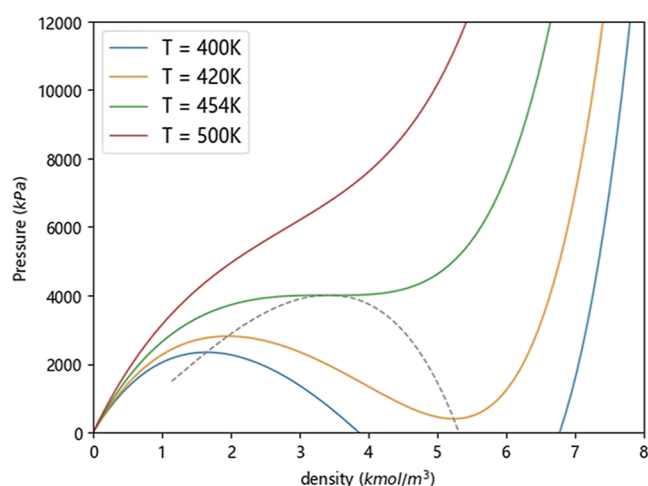


Figure 1. Pressure–molar density diagram for an equimolar mixture of ethane and *n*-heptane, generated using the PC-SAFT EoS.⁵ The solid lines depict the P – ρ isotherms: the 500 K curve corresponds to a mechanically supercritical isotherm, the 454 K line represents the critical isotherm, and the remaining curves are mechanically subcritical isotherms. The dashed curve represents the mechanical spinodal, whose vertex corresponds to the mechanical critical point, denoted by P_{mc} and ρ_{mc} , and marked by an asterisk in the figure.

a given pressure. While more complex root distributions are possible (e.g., with advanced EoS like PC-SAFT), the methodology presented herein focuses on the classical case of a single van der Waals loop.³⁶ In the case of three roots, the vapor phase selects the root with the largest compressibility factor (the smallest density root), while the liquid phase selects the root with the smallest compressibility factor (the largest density root), with the middle root. As the temperature increases, the isotherm tends to become linear, and the stability limits of the vapor and liquid phases gradually converge, eventually coinciding at the mechanical critical temperature. This point, as reiterated, is termed the mechanical critical point, where the mechanical stability limits of the vapor and liquid phases are enclosed by the spinodal curve. When the temperature reaches or exceeds the critical temperature, the extrema disappear and the equation of state has only one root. As the temperature continues to rise, the inflection point of the isotherm shifts progressively to the left, disappearing once the Boyle temperature is reached.

2.3. Density Extrapolation Strategy. Since the system within the mechanical spinodal curve is in an unstable state, a density extrapolation model can replace the EoS root, thereby accelerating the convergence of the phase equilibrium algorithm. Mathias et al.²¹ developed a density extrapolation strategy based on the mechanical spinodal curve, defining extrapolated liquid and vapor phase densities using [eqs 6a and 6b](#)

$$\rho^L = \exp\left(\frac{P - k_0^L}{k_1^L}\right) + 0.7\rho_{mc} \quad (6a)$$

$$\rho^V = \rho^{VB} + \frac{-k_1^V - \sqrt{(k_1^V)^2 - 4k_2^V(k_0^V - \frac{1}{P})}}{2k_2^V} \quad (6b)$$

where the superscript VB denotes the extrapolated boundary for the vapor phase, with L and V representing the liquid and vapor phases, respectively. The coefficients k_0^V , k_1^V , and k_2^V are the extrapolation parameters. The densities extrapolated by [eqs 6a and 6b](#) are continuous and smooth. The criteria region for the extrapolation function is empirically defined, with the boundaries of extrapolation criteria given by Mathias²¹ and others as follows

$$P_\rho \leq 0.1RT \stackrel{\text{def}}{=} P_\rho^{iB}, \quad i = L \text{ or } V \quad (7a)$$

$$\rho^L \leq \rho_{mc} \stackrel{\text{def}}{=} \rho(T_{mc}, P^{LB}, z) \quad (7b)$$

where P_ρ denotes the first-order partial derivative of pressure with respect to density, and P_ρ^{iB} is the first-order derivative at the extrapolated boundary of phase *i* (L for liquid, V for vapor). ρ^L is the liquid phase density; T and T_{mc} are the system temperature and the mechanical critical temperature, respectively; and R is the universal gas constant. The boundary of extrapolation criteria is defined when equality holds in [eqs 7a and 7b](#), which is denoted as the extrapolation boundary or extrapolation starting point.

[Figure 2](#) illustrates the variation in the solution of [eq 7a](#) with the temperature. As the temperature increases, the number of

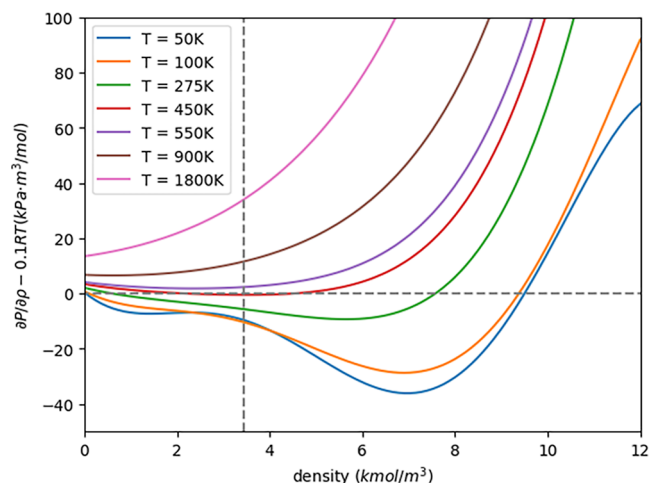


Figure 2. Behavior of the function $F(\rho) = \partial P / \partial \rho - 0.1RT$ for the equimolar ethane/*n*-heptane mixture represented by the PC-SAFT EoS. The vertical dashed line represents the mechanical critical density for this system.

intersection points (horizontal dashed line) between the solution curve and the isotherms gradually decreases from 2 to 0. When there are 2 or 1 intersection points, both the vapor and liquid phases use eq 7a as the extrapolation criteria. When there are 0 intersection points, the vapor phase adopts the EoS solution, while the liquid phase uses eq 7b as the extrapolation criteria. For the liquid phase, ρ_L satisfying eq 8 always exists in the mechanically subcritical state and may also exist in the mechanically supercritical state (since P_ρ^{LB} is slightly greater than zero). If eq 8 is not satisfied, then eq 7b is used as the extrapolation criteria

$$P_{\rho^L} \leq P_\rho^{\text{LB}} \quad (8)$$

The pressure at the extrapolation starting point, P and P_ρ^{LB} , needs to satisfy the equation of state, from which we can obtain the extrapolation coefficients in eqs 6a and 6b. For the vapor phase extrapolation, the coefficients must also satisfy the condition that the reciprocal of pressure tends to zero as the vapor phase density approaches the average of the extrapolated vapor density and the mechanical critical density, i.e.

$$\lim_{\rho \rightarrow \frac{\rho^{\text{VB}} + \rho_{\text{mc}}}{2}} \frac{1}{P(T, \rho^{\text{V}}, z)} = 0 \quad (9)$$

Match with the above condition, k_0^{L} and k_1^{L} are calculated by

$$k_0^{\text{L}} = P^{\text{LB}} - k_1^{\text{L}} \ln(\rho^{\text{LB}} - 0.7\rho_{\text{mc}}) \quad (10a)$$

$$k_1^{\text{L}} = P_\rho^{\text{LB}}(\rho^{\text{LB}} - 0.7\rho_{\text{mc}}) \quad (10b)$$

And k_0^{V} , k_1^{V} , and k_2^{V} are calculated by

$$k_0^{\text{V}} = \frac{1}{P^{\text{VB}}} \quad (11a)$$

$$k_1^{\text{V}} = -\frac{P_\rho^{\text{VB}}}{(P^{\text{VB}})^2} \quad (11b)$$

$$k_2^{\text{V}} = -\frac{\left| k_0^{\text{V}} + k_1^{\text{V}} \left(\frac{\rho_{\text{mc}} - \rho^{\text{VB}}}{2} \right) \right|}{\left(\frac{\rho_{\text{mc}} - \rho^{\text{VB}}}{2} \right)^2} \quad (11c)$$

eqs 11a–11c represent the coefficients used by Watson and Barton. The absolute value in eq 11c provides a mathematical guarantee for the sign of k_2^{V} .

Notably, Zhao and Saha²² interpreted the vapor phase model proposed by Mathias et al.²¹ in the form of eq 12. Furthermore, Xu et al.³⁷ proposed a more complex extrapolation model in their patent, in which the vapor phase extrapolation point depends on the results of the liquid phase extrapolation. This approach leads to a higher computational cost when applied to noncubic EoS lacking closed-form solutions

$$\rho^{\text{V}} = k_0^{\text{V}} + \frac{k_1^{\text{V}}}{P} + \frac{k_2^{\text{V}}}{P^2} \quad (12)$$

Although extrapolation functions may take various forms and are not strictly constrained by the EoS,²¹ care must be taken to avoid discontinuities in pseudoproperties or numerical singularities. While eqs 6a and 6b are mathematically simple and ensure smooth continuity, their applicability to noncubic EoS remains limited. This is primarily because the mechanical

critical condition ρ_{mc} has an analytical solution only for CEoS, whereas noncubic models lack such solutions, making direct numerical solving prone to instability. To overcome these challenges, this study proposes a new extrapolation function for the vapor phase, along with improved methods for determining the ρ_{mc} during the search for the liquid phase and a method for simultaneously searching the density and extrapolated boundary. These advancements enhance the efficiency and stability of extrapolation in noncubic EoS.

3. METHOD

The effectiveness and efficiency of the density extrapolation method depend on three primary factors: (1) the choice of algorithm for finding the appropriate pseudodensity; (2) the timing of when the pseudodensity method is initiated; and (3) how phase densities are accurately distinguished. This study proposes a systematic strategy that focuses on the three aspects mentioned above, tailored to the characteristics of noncubic EoS. First, an algorithm is used that simultaneously searches for both the EoS solution and the extrapolation starting point, ensuring consistency in the process and making effective use of the available information. Second, a hybrid algorithm combining linear extrapolation and Newton's method is employed to stabilize the solution for mechanical critical conditions. Finally, a more stable vapor phase density extrapolation method is introduced to prevent the algorithm from falling into the negative pressure region. The complete pseudocode for the algorithm is shown in algorithm new density extrapolation method (NDEM) and S1–S5 of the Supporting Information.

3.1. Density Solution and Extrapolation Boundary Search Method. The algorithm for simultaneously solving the EoS and identifying the extrapolation starting point is based on the second-order Taylor expansion of the pressure–density function

$$P^\Delta = P^k + \left(\frac{\partial P}{\partial \rho} \right)_{\rho=\rho^k} \delta^k + \frac{1}{2} \left(\frac{\partial^2 P}{\partial \rho^2} \right)_{\rho=\rho^k} (\delta^k)^2 + o((\delta^k)^2) \quad (13)$$

$$\delta^k = \rho^{k+1} - \rho^k \quad (14)$$

where P^Δ represents the target pressure, and P^k , δ^k , and $o((\delta^k)^2)$ are the pressure, iteration step, and Peano remainder for the k -th iteration, respectively. Neglecting the Peano remainder, eq 13 can be treated as a quadratic equation in δ^k .

Let $P_\rho^k = \left(\frac{\partial P}{\partial \rho} \right)_{\rho=\rho^k}$, and $P_{\rho\rho}^k = \left(\frac{\partial^2 P}{\partial \rho^2} \right)_{\rho=\rho^k}$. The root-finding formula yields

$$\delta^k = \Xi^k \pm \Theta^k \quad (15)$$

$$\Xi^k = -\frac{P_\rho^k}{P_{\rho\rho}^k} \quad (16)$$

$$\Theta^k = \frac{\sqrt{(P_\rho^k)^2 - 2P_{\rho\rho}^k(P^k - P^\Delta)}}{P_{\rho\rho}^k} \quad (17)$$

Equation 15–17 are the Newton–Raphson update algorithms that incorporate second-order derivative information. This second-order method effectively combines a first-order prediction of the extrema points with a prediction of the

parabolic deviation. When the square root term in eq 17 leads to an imaginary root, Θ^k takes zero, as shown in eq 18. Thus, eq 15 is simplified into a first-order prediction

$$\Theta^k = \frac{\sqrt{\max(0, (P_\rho^k)^2 - 2P_{\rho\rho}^k(P^k - P^\Delta))}}{P_{\rho\rho}^k} \quad (18)$$

Mills et al.²⁷ first applied eqs 15 and 16 and eq 18 to solving for density and extrema points. To integrate the extrapolated density approach, we modify eq 16 as follows

$$\Xi^k = -\frac{P_\rho^k - g(\Theta^k)P_{\rho\rho}^{\text{IB}}}{P_{\rho\rho}^k}, \quad i = \text{L or V} \quad (19)$$

where $g(x)$ is a nonsmooth function defined as follows

$$g(x) = \begin{cases} 0, & \text{if } x > 0 \\ 1, & \text{if } x \leq 0 \end{cases} \quad (20)$$

A phase-consistent solution refers to a solution in which the density is consistent with the respective phase (liquid or vapor). If no phase-consistent solution exists, eq 19 triggers the search for the extrapolation starting point. If a phase-consistent solution exists and it is at a certain distance from the extrapolation point, eqs 14 and 15 and 18 and 19 are applied to ensure detection of solutions prior to the β -sign change. Otherwise, a backtracking step is performed to locate the point where the step had been exceeded. In our approach, this is implemented by using first-order prediction only when conditions $(\partial P/\partial \rho_i)^k < P_\rho^{\text{IB}}$ and $i(P^k - P) < 0$ are satisfied, where i denotes the phase (with $i = 1$ for vapor and $i = -1$ for liquid). It is important to note that the initial iterative density ρ^0 should be physically meaningful, selected in the region of $\beta > 0$ and $V > b$, and be sufficiently low for the vapor phase or sufficiently high for the liquid phase.

A primary challenge is to locate a predefined starting point for extrapolation, even when no physical root exists for the phase of interest. A pure second-order Newton method, while efficient, can converge to nonphysical, complex roots in such cases (as evident from the radical expression in eq 18). To overcome this limitation, we developed a hybrid approach. Our method applies a second-order update but switches to a first-order Newton step when no real root is detected, ensuring a feasible starting point. The key idea of witching between root-finding and extrapolation is general and not tied to Newton's method, though we use it for clarity. The framework is also compatible with other iterative schemes such as Halley's method.

When the P - ρ isotherm exhibits a flat region, since the value of P_ρ^k is approximately zero, the step prediction δ in eq 14 may overshoot and requires correction. Such flat regions generally occur near the mechanical critical conditions. For numerically noncubic EoS, flat regions may also appear under low-temperature conditions. As P_ρ^k approaches zero in these regions, eq 14 may overestimate step δ , necessitating a correction. Mills et al.²⁷ used a fixed step α in place of δ , but this method can be overly conservative. We propose a hybrid approach that combines relative steps $\gamma\delta$ (with γ as an empirical constant) and fixed steps α

$$\delta = \text{sgn}(\delta) \cdot \min(\gamma\delta, \alpha) \quad (21)$$

where $\text{sgn}(x)$ is a nonsmooth sign function, defined as

$$\text{sgn}(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases} \quad (22)$$

eqs 21 and 22 ensure that when the current point is far from the solution point, a fixed step is used, while a relative step is applied when the solution is closer. The simultaneous density-boundary search algorithm is outlined in the pseudocode of Algorithm S2.

3.2. Density Extrapolation Models and Mechanical Critical Density Solution. **3.2.1. Mechanical Critical Density Solution.** The liquid phase extrapolation model ($i = \text{liquid}$) is presented in Algorithm S4, where the mechanical critical density is critical for the density extrapolation. However, for noncubic EoS, ρ_{mc} does not have an analytical solution and requires numerical methods. Watson and Barton²⁵ used a weighted approach for estimating the critical parameters of pure substances

$$\rho_{\text{mc}}^0 = \sum_{i=1}^n x_i \rho_{c,i} \quad (23a)$$

$$T_{\text{mc}}^0 = \sum_{i=1}^n x_i T_{c,i} \quad (23b)$$

However, this linear weighting method for critical parameters overlooks the system's nonidealities, and for complex substances like polymers, critical parameters of pure components are not easily accessible. Additionally, several studies^{38–40} have reported that EoS with high numerical complexity (such as the PC-SAFT EoS) may produce spurious mechanical critical points at low temperatures, which adds extra challenges to the numerical solution.

To address this, we propose a hybrid algorithm for mechanical critical conditions ($T_{\text{mc}}, \rho_{\text{mc}}$), presented in Algorithm S5. The algorithm first separates the iterative variables for linear extrapolation and then refines the solution using a trust-region Newton-like method. As illustrated in Figure 3, the algorithm starts at an initial point A at a fixed temperature, solves the inflection point equation to reach point B, and obtains derivative information. The information is then used to

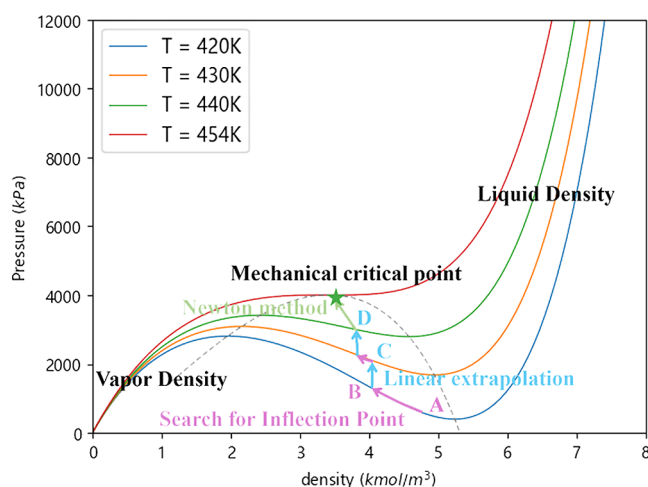


Figure 3. Search procedure for the mechanical critical point using the hybrid algorithm, with initial point A on the 420 K isotherm from Figure 1 as an example.

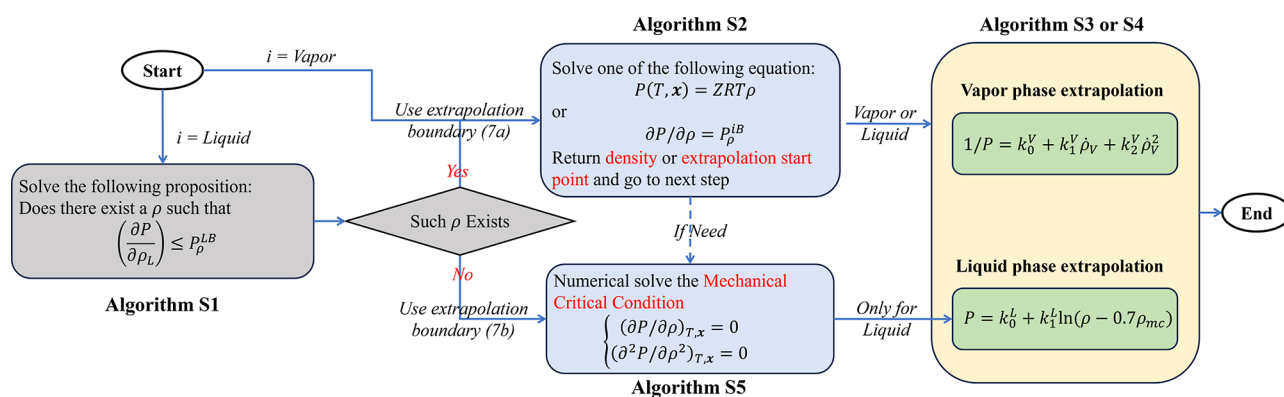


Figure 4. Workflow of the new density extrapolation method.

linearly extrapolate the current state, yielding the new state point C. This process is repeated for each new state point, while the iterative state points gradually approach the mechanical critical point. After several iterations of linear extrapolation, the Newton method with a trust-region strategy is applied from point D to find the final mechanical critical solution. The hybrid algorithm is capable of obtaining solutions over a wide range and offers higher efficiency and robustness compared to the trust-region Newton-like method alone.

It is important to note that the physical inflection point solution of the P – ρ isotherm exists only below the Boyle temperature. To avoid nonphysical, inflection point solutions, the iteration temperature can be reduced as needed. Furthermore, if eq 8 is satisfied and the verification process finds that the prescribed pressure is lower than the extrapolated pressure, then the search process for eq 8 can be restarted as the initial point for the numerical scheme described above. This approach accelerates the extrapolation in the mechanically subcritical region, improving the efficiency of routine applications.

3.2.2. Vapor Phase Extrapolation Model. The vapor phase extrapolation model ($i = \text{vapor}$) is illustrated in Algorithm S3. Equation 11c defines the extrapolation limit as the average of the extrapolated vapor density ρ^{VB} and the mechanical critical density ρ_{mc} . Watson and Barton²⁵ approximated ρ_{mc} when the EoS cannot be analyzed, using a weighted average of the pure component critical densities and temperatures. However, this approach depends heavily on the accuracy of the critical parameters and may lead to unphysical predictions such as negative pressures, especially for certain systems at low temperatures.

To address this, we propose a simple and practical correction: we define the extrapolation limit as $\alpha\rho^{\text{VB}}$, instead of using the average of ρ_{mc} and ρ^{VB} . In cases where the system temperature is below the mechanical critical temperature, the condition $\rho^{\text{VB}} \ll \rho_{\text{mc}}$ typically holds. For systems exhibiting a wide negative pressure region, the vapor phase region is typically narrow and thus $\rho^{\text{VB}} \ll \rho_{\text{mc}}$. Using $\alpha\rho^{\text{VB}}$ as a more conservative alternative to $0.5(\rho_{\text{mc}} + \rho^{\text{VB}})$, it can avoid entering the negative pressure region. Moreover, ρ^{VB} is only weakly temperature-dependent and remains significantly lower than ρ_{mc} even near the mechanical critical temperature.

The empirical factor α ranges from 1.2 to 2.0, with 1.5 recommended and proven effective in our extensive testing. This range is determined based on the following two facts: (a) the negative pressure region occurs only under subcritical

conditions, where a narrow van der Waals loop exists between the mechanical instability limits of the liquid and vapor phases. (b) The second derivative of pressure, with respect to density, is often nearly linear. Consequently, the trend of the first derivative at the vapor starting point ρ^{VB} can be used to infer that the left boundary of the negative pressure region is approximately at $2\rho^{\text{VB}}$. Further, a lower bound is necessary, as it should be higher than the extrapolation starting point itself.

Accordingly, eq 11c is reformulated as eq 24

$$k_2^{\text{V}} = -\frac{[k_0^{\text{V}} + k_1^{\text{V}}(\alpha - 1)\rho^{\text{VB}}]}{(\alpha - 1)^2(\rho^{\text{VB}})^2} \quad (24)$$

which not only avoids the occurrence of negative pressure but also eliminates the dependence on the mechanical critical density ρ_{mc} , simplifying the calculation process while still providing a reasonable vapor extrapolation density.

3.2.3. New Density Extrapolation Method. The proposed new density extrapolation method for noncubic EoS is shown by algorithm NDEM and Figure 4, which consists of > Algorithms S1–S5. For vapor phase density solutions, the algorithm starts from the simultaneous density-boundary search algorithm (Algorithm S2). For liquid phase density solutions, the extrapolation boundary is first determined through a liquid phase extrapolation criterion identification algorithm (Algorithm S1). Algorithm S1 includes the procedure to verify whether there exists a density satisfying eq 7a. Generally, eq 7a is used for temperatures below the mechanical critical temperature, while eq 7b is used for temperatures above the mechanical critical temperature. If eq 7a is used as the extrapolation criteria, then Algorithm S2 is employed to simultaneously search for the density solution or the new extrapolation starting point. If the extrapolation starting point is returned, then the liquid phase extrapolation model from Algorithm S4 is applied to obtain the extrapolated density solution. Otherwise, the density solution from EoS is used directly. When the mechanical critical solution is required, Algorithm S5 is used to obtain the numerical solution.

The obtained values of ρ^{L} or ρ^{V} can be used to calculate other volumetric properties, such as deviations in enthalpy, entropy, and fugacity coefficients. In this work, the scaled fugacity coefficient²¹ $\phi_i^{\text{L}} = \phi_i^{\text{L}} \left(\frac{P}{P^{\text{L}}} \right)$ is used to obtain more reasonable pseudoliquid phase fugacity coefficients.

Table 1. Initial Guesses from Linear Extrapolation for System 1 at Different Temperatures and the Number of Iterations Required to Achieve the Desired Precision of the Solution

T [K]	T_{guess} [K]	ρ_{guess} [mol/m ³] ^a	Iter	T [K]	T_{guess} [K]	ρ_{guess} [mol/m ³] ^a	Iter
1475.01	1586.36	0.354	4	1757.70	1585.56	0.354	4
1495.20	1586.67	0.354	4	1777.89	1585.17	0.354	4
1515.39	1586.90	0.354	4	1798.08	1584.73	0.354	4
1535.59	1587.08	0.354	3	1818.27	1584.27	0.355	5
1555.78	1587.20	0.354	3	1838.46	1583.76	0.355	5
1575.97	1587.25	0.354	3	1858.66	1583.23	0.355	5
1596.16	1587.24	0.354	3	1878.85	1582.67	0.355	5
1616.35	1587.04	0.354	3	1899.04	1582.08	0.355	5
1636.55	1586.63	0.354	4	1919.23	1581.45	0.355	5
1656.74	1586.02	0.354	4	1939.42	1580.81	0.355	5
1676.93	1586.76	0.354	4	1959.62	1580.13	0.355	5
1697.12	1586.53	0.354	4	1979.81	1579.43	0.355	5
1717.31	1586.25	0.354	4	2000.00	1578.71	0.355	5
1737.51	1585.92	0.354	4				

^aThe mechanical critical molar density is represented in terms of the polymer's long-chain molecular weight.

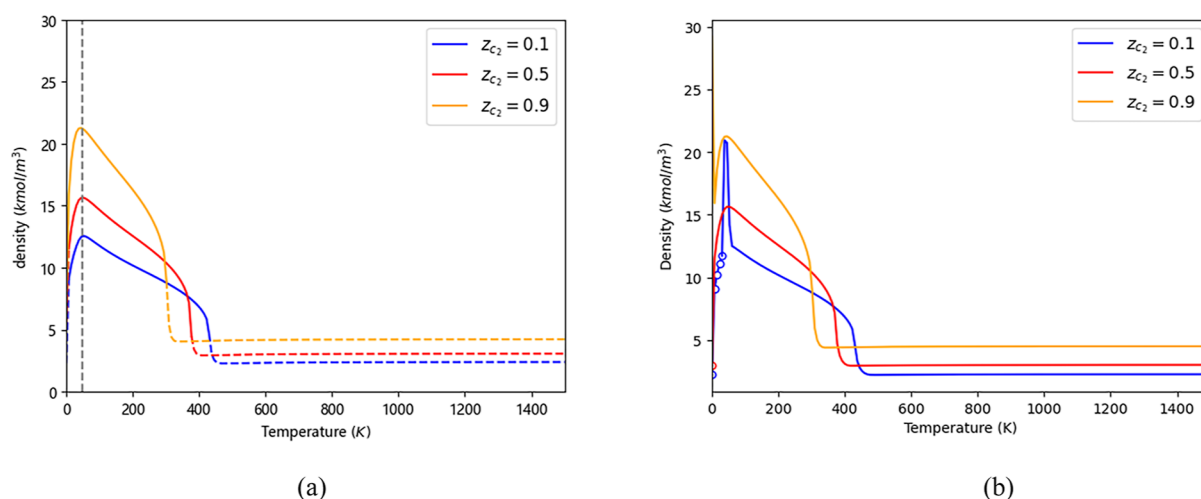


Figure 5. Variation of liquid phase density with temperature for three ethylene/*n*-butene binary mixtures with different compositions at 0.1 MPa. (a) Results from the method proposed in this study, with the dashed line representing the extrapolated liquid phase density. (b) Results from the commercial process simulator Aspen Plus v11, where hollow circles indicate data points where Aspen Plus v11 encountered errors.

4. RESULTS AND DISCUSSION

To validate the accuracy and efficiency of the proposed method, six case studies are presented. All case studies are constructed using the PC-SAFT EoS, which features strong nonlinearity and significant numerical complexity. The pure component parameters for all components can be found in Tables S2 and S3 of the [Supporting Information](#). Unless otherwise specified, vapor phase results are computed using [eq 24](#) with the empirical factor $\alpha = 1.5$:

Case 1: Comparison of the stability between the hybrid algorithm and the standalone Newton method in determining mechanical critical conditions.

Case 2: Analysis of the continuity and smoothness of vapor and liquid densities across a wide temperature range.

Case 3: Application of the method to flash calculations and dew/bubble point estimations for multicomponent systems.

Another three cases highlighting the capabilities of avoiding the negative pressure and nonsmoothness of extrapolated pseudoproperties are available in the [Supporting Information](#).

4.1. Case 1: Comparison of Hybrid and Newton Methods. To compare the impact of [Algorithm S5](#) with a

direct newton-like method, we conduct liquid phase density extrapolation for **System 1**, whose composition is provided in Table S2 of the [Supporting Information](#). A high-complexity working range is selected to increase the likelihood of triggering the extrapolation module. The pressure is fixed at 1 Pa, with 100 uniformly spaced temperature points sampled from 1 to 2000 K. A quasi-Newton method is employed with a convergence tolerance of 10^{-4} and a maximum of 50 iterations.

According to [Algorithm S1](#), for the liquid phase:

1. When $T < 1475.01$ K, the partial derivative $P_{\rho L}$ confirms a mechanically subcritical state, and [Algorithm S2](#) yields a valid density solution.
2. When $1475.01 \leq T < 1676.93$ K, [Algorithm S2](#) identifies the extrapolation boundary, activating both the density extrapolation module and the mechanical critical condition solver.
3. When $T > 1676.93$ K, [Algorithm S1](#) detects that the extrapolation boundary defined by [eq 7a](#) is no longer feasible and thus directly computes the mechanical critical condition and corresponding extrapolated density.

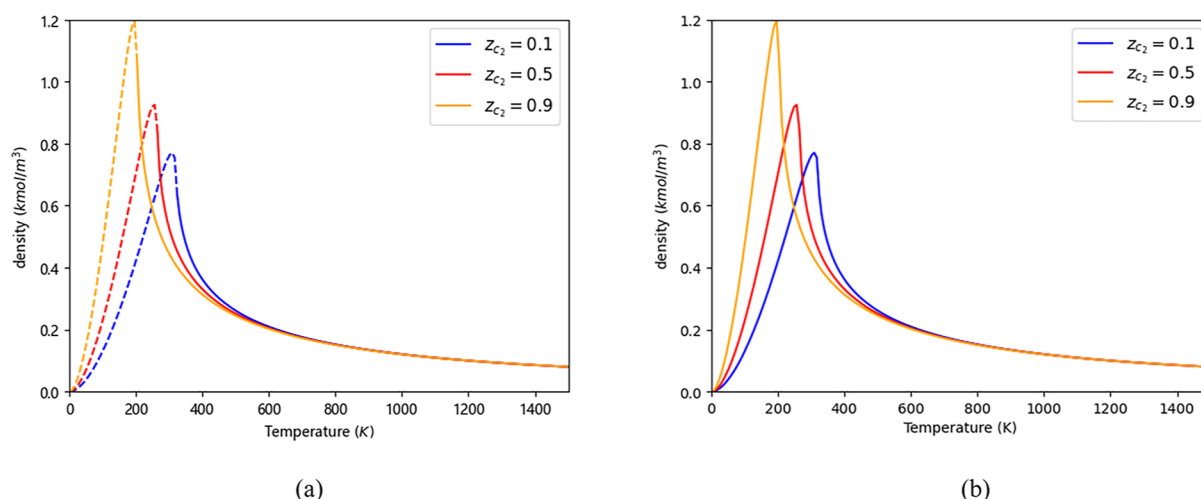


Figure 6. Variation of vapor phase density with temperature for three ethylene/n-butene binary mixtures with different compositions at 1 MPa. (a) Results from the method proposed in this study, with the dashed line representing the extrapolated vapor phase density. (b) Results from the commercial process simulator Aspen Plus v11.

In the absence of prior data, the search for the mechanical critical point must rely on heuristic or empirical initial guesses. This typically leads to a poor performance and low efficiency. To evaluate this, we compare two initialization strategies:

1. A mole fraction weighted average of pure component critical parameters (from the Aspen Plus v11 database; Tables S2 and S3).
2. Linear extrapolation from known conditions.

Among the 100 selected temperature points, 73 points that do not require extrapolation are discarded, leaving 27 points that require extrapolation.

Using Algorithm S5, the correct results are obtained for all 27 sets: $(T_{\text{mc}}, \rho_{\text{mc}}) = (1587.26 \text{ K}, 0.35397 \text{ mol/m}^3)$. Table 1 presents the linear extrapolation results $(T_{\text{guess}}, \rho_{\text{guess}})$ for each set along with the number of iterations (Iter) of the quasi-Newton method based on these results as initial guesses. As shown, the linear extrapolation results are close to the true values, allowing the quasi-Newton method to converge within 5 iterations. In contrast, the molar weighting method based on pure component parameters provides the same initial guess for all 27 sets: $(T_{\text{guess}}, \rho_{\text{guess}}) = (548.35 \text{ K}, 47.899 \text{ mol/m}^3)$. Consequently, the subsequent quasi-Newton method fails to obtain the mechanical critical solution due to the loss of the descent direction during iteration.

4.2. Case 2: Continuity in Phase Densities for Binary Mixtures. This case examines three binary ethylene/n-butene mixtures (Systems 2–4) with different compositions. For each system, liquid phase densities at 0.1 MPa and vapor phase densities at 1 MPa are computed across a temperature range of 0.5–1500 K. The results obtained using the proposed method are compared with those from the commercial simulator Aspen Plus v11, as shown in Figures 5 and 6.

Figures 5a and 6a demonstrate that the liquid and vapor phase density isobars generated by the present method are continuous across the temperature domain. In Figure 5a, the liquid phase isobar is split into two regions by a vertical dashed line. To the left of the dashed line, Aspen Plus version 11 encounters runtime errors, which are denoted by circles in Figure 5b. This issue arises from the complexity of the second Maxwell loop^{36,41} of the PC-SAFT EoS at low temperatures.

Under mechanically supercritical conditions or above the Boyle temperature, the liquid phase density is replaced by extrapolated values that approach the extrapolation limit, resulting in a nearly flat trend, as seen in Figure 5a. For the vapor phase, where no EoS solution exists at low temperatures, the density is extrapolated instead. As shown in Figure 6, these extrapolated values agree well with those predicted by Aspen Plus v11.

To further test the algorithm's robustness, a more complex natural gas mixture (System 5) used by Watson and Barton²⁵ is simulated. The mixture contains 1.00% N_2 , 91.60% CH_4 , 4.93% C_2H_6 , 1.71% C_3H_8 , 0.35% $n\text{-C}_4\text{H}_{10}$, 0.40% $i\text{-C}_4\text{H}_{10}$, and 0.01% $i\text{-C}_5\text{H}_{12}$ (mole fractions) and is modeled at 5.0 MPa. As shown in Figure 7, at low temperatures, no high-pressure EoS

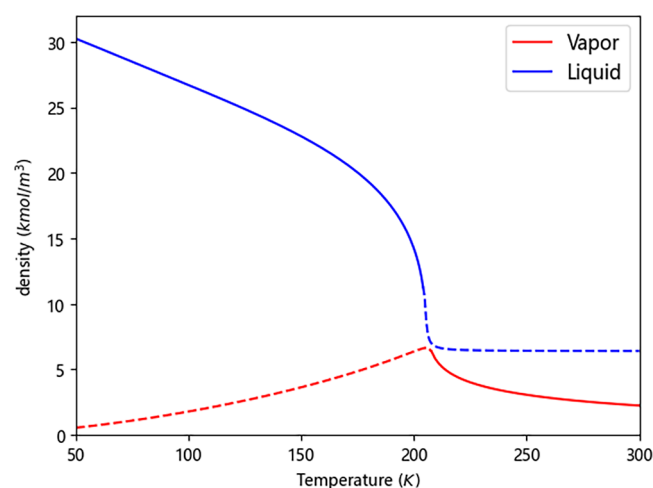


Figure 7. Temperature dependence of the liquid and vapor phase molar densities for System 5 at 5.0 MPa.

solution exists for the vapor phase; therefore, the vapor density is extrapolated. At high temperatures, no low-pressure EoS solution exists for the liquid phase, prompting extrapolation. Near the segmentation point, the vapor phase density closely matches the liquid phase density due to the temperature approaching the mechanical critical temperature and the pressure nearing the mechanical critical pressure.

Table 2. Comparison of PT Flash Algorithm with and without Extrapolation for Systems 5–7^a

substance system	extrapolation status	<i>T</i> [K]	<i>P</i> [MPa]	VFRAC	ρ^L [mol/m ³]	ρ^V [mol/m ³]
System 5	without extrapolation	295.15	5.5	1	2568.7704	2568.7703
	with extrapolation			1	6225.1737	2568.7703
	without extrapolation	204.71	5.5	0.4773	14952.4990	7407.4142
	with extrapolation			0.4773	14952.5012	7407.4148
	without extrapolation	110.15	5.5	0	25988.9737	25988.9737
	with extrapolation			0	25988.9737	2621.5274
System 6	without extrapolation	550.00	5.0	0	1454.4244	1450.4408
	with extrapolation			0.9963	8635.4505	1450.1686
	without extrapolation	450.00	1.0	0.4611	297.3199	296.1216
	with extrapolation			0.9930	3130.6753	295.0697
System 7	without extrapolation	370.00	0.3	0	99.0848	99.0848
	with extrapolation			1	3332.4902	99.0848
	without extrapolation	250.00	0.5	0.6086	11606.8437	251.8171
	with extrapolation			0.6086	11606.8437	251.8171
	without extrapolation	100.00	3.0	1	18749.9780	18749.9782
	with extrapolation			0	18749.9782	3774.6435

^aThe symbol VFRAC represents the mole fraction of the vapor phase in the feed stream.

Table 3. Comparison of the Bubble/Dew Point Pressure Algorithm with and without Extrapolation for System 8^a

substance system	extrapolation status	<i>T</i> [K]	VFRAC	<i>P</i> _{guess} [MPa]	ρ^L [mol/m ³]	ρ^V [mol/m ³]	<i>P</i> [MPa]	θ_A
$\frac{n_A}{n_B}$: $\frac{0.27}{0.73}$ methane/ <i>n</i> -butane (System 8)	without extrapolation	394.15	0	5.0	3271.7703	2969.2373	4.97	0.290
	with extrapolation				6596.3311	3955.8909	7.13	0.426
	without extrapolation	409.00	1	2.0	685.4677	685.4663	2.00	0.270
	with extrapolation				5000.4657	4468.4138	6.51 ^b	0.244

^aThe symbol VFRAC represents the mole fraction of the vapor phase in the feed stream, *P*_{guess} represents the initial guessed pressure, and θ_A represents the mole fraction of the first component in the liquid or vapor phase obtained from the solution. ^bThe branching solution for retrograde condensation.

Table 4. Comparison of the Bubble/Dew Point Temperature Algorithm with and without Extrapolation for Systems 8^a

substance system	extrapolation status	<i>P</i> [MPa]	VFRAC	<i>T</i> _{guess} [K]	ρ^L [mol/m ³]	ρ^V [mol/m ³]	<i>T</i> [K]	θ_A
$\frac{n_A}{n_B}$: $\frac{0.80}{0.20}$ ethane/ <i>n</i> -decane (System 9)	without extrapolation	10.0	0	350	9185.4145	9185.4145	363.25	0.800
	with extrapolation				8426.4957	5393.1809	387.74	0.971
	without extrapolation	10.0	1	350	2281.2747	2281.1101	620.16	0.800
	with extrapolation				4900.0097	3602.5929	510.45	0.632

^aThe symbol VFRAC represents the mole fraction of the vapor phase in the feed stream, *T*_{guess} represents the initial guessed temperature, and θ_A represents the mole fraction of the first component in the liquid or vapor phase obtained from the solution.

4.3. Case 3: Flash and Bubble/Dew Point Calculations for Multicomponent Systems. PT flash and bubble and dew point calculations are widely used in thermodynamic modeling. PT flash solves for phase compositions and fractions given the temperature, pressure, and overall composition. The bubble point and dew point problems are the specialized case of the TV flash and PV flash problems, respectively. They are designed to calculate the bubble (dew) point pressure or bubble (dew) point temperature at a specified temperature or pressure.

Three mixtures are considered for PT flash:

1. **System 5:** a complex natural gas mixture of Case 2.
2. **System 6:** a polymer containing mixture, whose composition is given in Table S2 of the [Supporting Information](#).
3. **System 7:** The refrigerant stream used by Watson and Barton,²⁵ with a composition of 5.82% N₂, 20.62% CH₄, 39.37% C₂H₆, and 34.19% *n*-C₄H₁₀.

Table 2 summarizes PT flash results under various temperature and pressure conditions with and without the

density extrapolation algorithm. Tables 3 and 4 present the results for bubble/dew point calculations for **System 8** and **System 9**. The results clearly show that without density extrapolation, a total of 10 examples out of 12 examples converge to trivial solutions. The trivial solution generally occurs near single-phase regions, which can also be demonstrated by the result that all bubble/dew point calculations without extrapolation fail. By introducing physically reasonable extrapolated densities that differentiate between phases, the algorithm allows the solver to bypass such regions and reach valid equilibrium solutions. Importantly, the method is fully applicable to polymer containing systems and supports robust flash calculations. In the two-phase region away from the phase boundaries, where both phases admit valid roots, our algorithm and conventional solvers avoid trivial solutions and produce identical results.

Although Stateva and Tsvetkov⁴² conducted a quantitative analysis of the convergence benefits of density extrapolation based on the contraction mapping principle,⁴³ such analyses are limited to fixed point methods and do not extend easily to

other classes of solvers. Therefore, a formal convergence analysis is not pursued in this work.

5. CONCLUSION

This work proposes a general density extrapolation algorithm for noncubic EoS. It comprises a liquid phase extrapolation criterion identification algorithm (Algorithm S1), a simultaneous density-boundary search algorithm (Algorithm S2), a modified vapor phase extrapolation method (Algorithm S3), a liquid phase extrapolation method (Algorithm S4), and a hybrid numerical solution for mechanical critical conditions (Algorithm S5).

Algorithm S2 integrates the traditional density solving process of the EoS with the density extrapolation method for detecting the extrapolation starting point, forming a complete solution framework. By utilizing process information, this approach eliminates the need to reexplore the extrapolation starting point when the density solution search fails. To address the negative pressure issue in density extrapolation, Algorithm S3 eliminates the reliance on mechanical critical conditions. Although conservative, it still provides reasonable vapor phase densities in strongly nonideal systems, such as polymers with narrow vapor phase regions and wide negative pressure zones.

Since no analytical expression exists for the mechanical critical conditions of higher-order EoS, we propose a robust hybrid algorithm (Algorithm S5) for numerically solving these conditions. The results demonstrate that it is more robust than using a pure component parameter weighting method, and its convergence domain is significantly more stable than that of the direct Newton method. The proposed algorithm can effectively address the numerical instability of the pressure-explicit EoS when the critical parameters of pure components are unavailable.

In conclusion, the method proposed in this article ensures reasonable and smooth phase density results through density extrapolation when the EoS yields unreasonable phase densities. While demonstrated using the PC-SAFT EoS, its primary value lies in providing a numerical stability framework for noncubic EoS that lacks a general analytical mechanical stability criterion. The generic algorithm design ensures extensibility to other noncubic EoS and potential applicability to cubic EoS.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.iecr.5c04076>.

Definition of thermodynamic stability of mixtures, supplementary cases, and pseudocode of this work (PDF)

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Notes

The authors declare no competing financial interest.

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■ NOMENCLATURE

Roman Letters

F feed molar flow (mol/s)
 H molar enthalpy (J/mol)
 k constant in extrapolation models
 K phase equilibrium constant
 L liquid phase molar flow (mol/s)
 N number of moles
 P absolute pressure (Pa)
 Q heat duty (J/s)
 R universal gas constant (J/(K·mol))
 T temperature (K)
 v molar volume (m³/mol)
 V volume (m³)
 z mixture mole fraction vector
 Z compressibility factor

Greek Letters

β isothermal compressibility
 γ constant in the relative step
 δ iteration step
 Θ second-order prediction
 κ corrector in the extrapolation liquid coefficient
 Ξ first-order prediction
 ρ molar density (mol/m³)
 ϕ fugacity coefficient

Subscripts

- A first component in a binary mixture
 c critical property
 i component indices
 mc mechanical critical property
 ρ partial derivative to molar density

Superscripts

- k iteration number
 L liquid phase property
 LB liquid extrapolation boundary
 V vapor phase property
 VB vapor extrapolation boundary
 Δ target value extrapolation property

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